YASSINE BEN SASSI

AI Software Engineer Intern | Data Engineering and Machine Learning

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student (Data Engineering and AI) who builds AI tools end to end: LLM and RAG agents measured against evaluation sets, PyTorch computer vision models, and Python and FastAPI services. Designed lakehouse and ETL pipelines that turn raw operational data into validated outputs for decision-making at a national energy company and a government ministry. Seeking an AI Software Engineer internship to experiment with open-weight and frontier models, evaluate them, and ship them into production tools.

TECHNICAL SKILLS

- AI and ML: PyTorch, Hugging Face, LLMs, RAG, fine-tuning, OpenCV, YOLO, scikit-learn, TensorFlow, MLflow, ONNX, PCA, feature engineering
- Software: Python, FastAPI, Django, Node.js, Express.js, NestJS, REST APIs, JavaScript, Java, C, Dart, Bash
- Data and Infrastructure: SQL, PostgreSQL, SQL Server, Apache Spark, Airflow, Kafka, dbt, ETL/ELT, Medallion lakehouse, Docker, Kubernetes, Git, CI/CD, Linux, AWS
- BI and Visualization: Power BI, Tableau, Streamlit, Plotly, Matplotlib

PROFESSIONAL EXPERIENCE

WMS Developer Intern | Axcess Logistics | Jul-Aug 2026

- Built backend services for a multi-tenant warehouse management system with NestJS, Prisma and PostgreSQL.
- Implemented role-based access control across 4 scoped roles, enforcing strict data isolation across warehouse and merchant boundaries.
- Added audit logging to every state-changing operation for traceability and compliance.

PunchTime Machine System Developer Intern | SCSI (Société de Conseils et Solutions Informatiques) | May-Jul 2025

- Developed a full-stack attendance, scheduling and history application for all company employees with .NET Core 8 and SQL Server.
- Designed the relational schema and an admin interface with role-based authentication.

Artificial Intelligence Intern | Idea Lab, FSB | Jul-Aug 2024

- Trained a conversational LLM agent to answer educational questions for preparatory-class students.
- Collected and structured the training corpus through web scraping and preprocessing.

SELECTED PROJECTS

Educational Conversational Agent (LLM + RAG) | Python, LLM, RAG

- Designed a RAG agent for course questions: over 90% relevance on manual evaluation and under 2 seconds average response time.
- Compared BM25, dense and hybrid retrieval, and selected the best-performing configuration for production.

Parkinson's Disease Detection from Voice Biomarkers | Python, scikit-learn, PCA

- Built a detection pipeline on 195 voice recordings (22 acoustic features) from 31 individuals; PCA reduced the features to 6 components.
- Benchmarked 5 classifiers with grid-search tuning and oversampling; the best model (MLP) reached 90% accuracy and 97% recall on positive cases.

Computer Vision MLOps Pipeline | Python, MLflow

- Automated training, versioning and deployment of a computer vision model with experiment tracking and a model registry.

Fridguard: AI-Powered Smart Refrigerator | YOLO, Node.js, MongoDB, Firebase, REST APIs

- Co-developed an edge-to-cloud pipeline: a Raspberry Pi 4 with camera and sensors sends data every 5 seconds to a Node.js hub.
- Orchestrated a Python YOLO service for food recognition, event-driven spoilage alerts, and REST APIs for an Android app.

STEG Energy Observatory: Intelligent Grid Management Platform | Python, ETL, Data Lakehouse, ML

- Designed a Bronze/Silver/Gold lakehouse and ETL pipelines over multi-year national energy production, incident and solar-subsidy data.
- Built Gold-layer KPIs for decision-makers, including a Gini coefficient for regional power-cut distribution, and contributed ML-based production forecasting.

EDUCATION

Mastère de Recherche, Internet des objets et analyse des données | École Polytechnique de Tunisie (EPT), University of Carthage | 2026-2027

Cycle Ingénieur, Computer Engineering | Faculté des Sciences de Bizerte, University of Carthage | 2024-2026

Integrated Preparatory Program | Faculté des Sciences de Bizerte, University of Carthage | 2022-2024

Baccalaureate, Computer Science | Tunisia | 2021-2022

CERTIFICATIONS

AWS Cloud Practitioner Essentials (AWS Skill Builder) | Introduction to Airflow in Python, Introduction to Docker, Understanding Data Engineering (DataCamp) | Web Development (GoMYCODE)

LANGUAGES

Arabic (native) | French (fluent) | English (intermediate) | German (beginner)